

Geometry

Unit 11

Lesson 9 Homework

Name _____

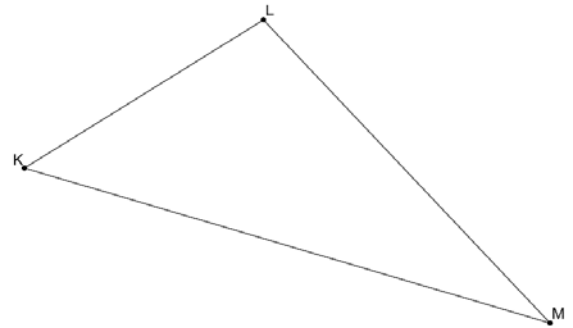
Date _____

Period _____

Use this information to answer questions 1 through 5.

Triangle KLM is shown, where

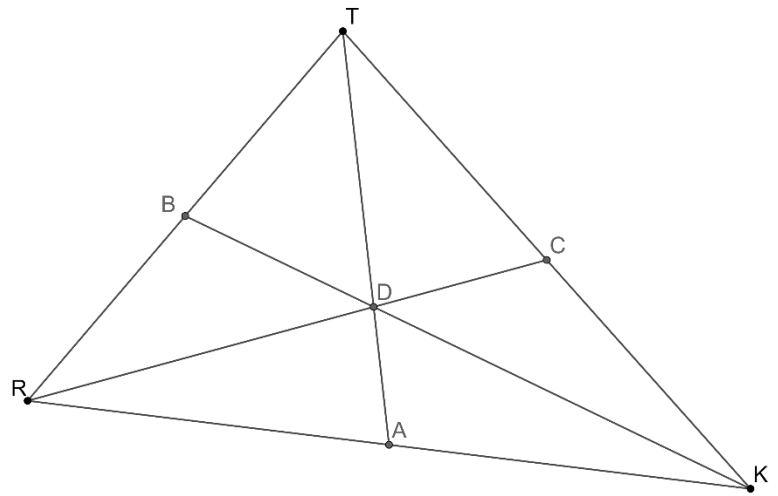
$m\angle K = (2x + 3)^\circ$, $m\angle L = (3x + 7)^\circ$, and $m\angle M = (x + 2)^\circ$.



1. Find the value of x .
2. Identify the measure of $m\angle K$.
3. Identify the measure of $m\angle L$.
4. Identify the measure of $m\angle M$.
5. List the sides of the triangle in order of their lengths, smallest to largest.

Use this information to answer questions 6 through 10.

Triangle RTK is shown, where $\overline{RC}$, $\overline{BK}$, and $\overline{AT}$ are median. $BT = 3$, $BK = 12$, $DC = 4$, $TD = 6$, and $RK = 16$.



6. Find the length of $\overline{BD}$.

7. Find the length of $\overline{RC}$.

8. Find the length of $\overline{DA}$.

9. Find the length of $\overline{RT}$.

10. Find the perimeter of triangle RBD .